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Charging inlet assembly having a proximity resistor assembly

Abstract

A charging inlet assembly includes a housing having a DC section with DC terminals received in DC terminal channels for mating with a DC charging connector and AC terminals, including a proximity terminal, received in AC terminal channels for mating with an AC charging connector. The charging inlet assembly includes an HMI assembly coupled to the rear of the housing having a control circuit board, a charging indicator coupled to the control circuit board visible from the front of the housing to indicate a charging status of the charging inlet assembly, and a proximity resistor assembly coupled to the housing. The proximity resistor assembly includes a resistor coupled to the control circuit board and a cable connector connected between the control circuit board and the proximity terminal.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application No. 63/234,668, filed 18 Aug. 2021, the subject matter of which is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

(1) The subject matter herein relates generally to charging inlet assemblies.

(2) Charging inlet assemblies are used to charge vehicles, such as for charging a battery system of an electric vehicle (EV) or hybrid electric vehicle (HEV). The charging inlet assembly includes power connectors for connecting to a charging connector. Conventional charging inlet assemblies include AC terminals and DC terminals. The AC terminals and the DC terminals are housed within an inlet housing, which interfaces with the charging connector. The inlet housing is coupled to the vehicle. Cable harnesses are coupled to the AC terminals and the DC terminals and extend to other system components, such as the battery. Standards for electrical connectors for electric vehicles, such as the SAE J1772 North American Standard, have signaling protocols to detect connection of the charging plug to the vehicle. For example, the charging inlet assembly typically includes a control circuit board retained within the housing. The control circuit board includes a proximity circuit having a resistor connected between the proximity signal and the ground. The housing includes a large pocket at the rear of the housing that receives the control circuit board. The pocket and the control circuit board increase the overall size, complexity and cost of the charging inlet

assembly.

(3) A need remains for an improved charging inlet assembly.

BRIEF DESCRIPTION OF THE INVENTION

(4) In one embodiment, a charging inlet assembly is provided and includes a housing extending between a front and a rear. The housing has a DC section includes DC terminal channels. The housing has an AC section includes AC terminal channels. The DC section configured for mating with a DC charging connector at the front. The AC section configured for mating with an AC charging connector at the front. The charging inlet assembly includes DC terminals received in the corresponding DC terminal channels for mating with the DC charging connector. The DC terminals include a proximity terminal. The charging inlet assembly includes AC terminals received in the corresponding AC terminal channels for mating with the AC charging connector. The charging inlet assembly includes a human machine interface (HMI) assembly coupled to the rear of the housing. The HMI assembly includes a control circuit board. The HMI assembly includes a charging indicator coupled to the control circuit board. The charging indicator is visible from the front of the housing to indicate a charging status of the charging inlet assembly. The charging inlet assembly includes a proximity resistor assembly coupled to the housing. The proximity resistor assembly includes a resistor coupled to the control circuit board. The proximity resistor assembly includes a cable connector connected between the control circuit board and the proximity terminal.

(5) In another embodiment, a charging inlet assembly is provided and includes a housing extending between a front and a rear. The housing has a DC section includes DC terminal channels. The housing has an AC section includes AC terminal channels. The housing has a human machine interface (HMI) section discrete from the DC section and the AC section. The DC section configured for mating with a DC charging connector at the front. The AC section configured for mating with an AC charging connector at the front. The housing includes a rear cavity at the HMI section. The rear cavity is separate from the DC terminal channels and from the AC terminal channels. The charging inlet assembly includes DC terminals received in the corresponding DC terminal channels for mating with the DC charging connector. The DC terminals include a proximity terminal. The charging inlet assembly includes AC terminals received in the corresponding AC terminal channels for mating with the AC charging connector. The charging inlet assembly includes an HMI assembly received in the rear cavity. The HMI assembly includes a control circuit board. The HMI assembly includes a charging indicator coupled to the control circuit board. The charging indicator located within the HMI section and is visible from the front of the housing to indicate a charging status of the charging inlet assembly. The charging inlet assembly includes a proximity resistor assembly coupled to the housing. The proximity resistor assembly includes a resistor coupled to the control circuit board. The proximity resistor assembly includes a cable connector connected between the control circuit board and the proximity terminal.

(6) In a further embodiment, a charging inlet assembly is provided and includes a housing extending between a front and a rear. The housing has a DC section includes DC terminal channels. The housing has an AC section includes AC terminal channels. The DC section configured for mating with a DC charging connector at the front. The AC section configured for mating with an AC charging connector at the front. The charging inlet assembly includes DC terminals received in the corresponding DC terminal channels for mating with the DC charging connector. The DC terminals include a proximity terminal. The charging inlet assembly includes AC terminals received in the corresponding AC terminal channels for mating with the AC charging connector. The charging inlet assembly includes a human machine interface (HMI) assembly coupled to the rear of the housing. The HMI assembly includes a control circuit board. The HMI assembly includes a charging indicator coupled to the control circuit board. The charging indicator is visible from the front of the housing to indicate a charging status of the charging inlet assembly. The charging inlet assembly includes a proximity resistor assembly coupled to the housing. The proximity resistor assembly includes a resistor and a cable connector. The resistor coupled to a proximity circuit of

the control circuit board. The resistor coupled to a ground of the control circuit board. The cable connector includes a contact connected to the proximity circuit of the control circuit board. The cable connector includes a cable connected between the contact and the proximity terminal to receive a proximity signal from the proximity terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front view of a charging inlet assembly including an AC charging module, a DC charging module, a human machine interface (HMI) assembly, and a proximity resistor assembly in accordance with an exemplary embodiment.
- (2) FIG. 2 is a rear view of the charging inlet assembly showing the AC charging module, the DC charging module, a human machine interface (HMI) assembly, and a proximity resistor assembly in accordance with an exemplary embodiment.
- (3) FIG. 3 is a rear perspective view of the charging inlet assembly showing the AC charging module, the DC charging module, a human machine interface (HMI) assembly, and a proximity resistor assembly in accordance with an exemplary embodiment.
- (4) FIG. 4 is a rear perspective view of a portion of the charging inlet assembly showing the proximity resistor assembly and the human machine interface (HMI) assembly in accordance with an exemplary embodiment.
- (5) FIG. 5 is a rear perspective view of a portion of the charging inlet assembly showing the proximity resistor assembly and a portion of the human machine interface (HMI) assembly in accordance with an exemplary embodiment.

DETAILED DESCRIPTION OF THE INVENTION

- (6) FIG. 1 is a front view of a charging inlet assembly **100** including an AC charging module **102**, a DC charging module **104**, a human machine interface (HMI) assembly **106**, and a proximity resistor assembly **200** in accordance with an exemplary embodiment. FIG. 2 is a rear view of the charging inlet assembly **100** showing the AC charging module **102**, the DC charging module **104**, a human machine interface (HMI) assembly **106**, and a proximity resistor assembly **200** in accordance with an exemplary embodiment. FIG. 3 is a rear perspective view of the charging inlet assembly **100** showing the AC charging module **102**, the DC charging module **104**, a human machine interface (HMI) assembly **106**, and a proximity resistor assembly **200** in accordance with an exemplary embodiment.
- (7) The charging inlet assembly **100** is used as a charging inlet for a vehicle, such as an electric vehicle (EV) or hybrid electric vehicle (HEV). The charging inlet assembly **100** is configured for mating reception with a charging connector (not shown). In an exemplary embodiment, the charging inlet assembly **100** is configured for mating with various types of charging connectors, such as a DC fast charging connector (for example, the SAE combo CCS charging connector) in addition to AC charging connectors (for example, the SAE J1772 charging connector).
- (8) The charging inlet assembly **100** includes a housing **110** configured to be mounted in the vehicle. The housing **110** holds the AC charging module **102** and the DC charging module **104** for mating with the charging connector. In various embodiments, the AC charging module **102** and/or the DC charging module **104** are removable from the housing **110**. For example, the charging modules **102** may be coupled to the housing **110** using latches, fasteners, clips, or other securing means. The charging modules **102**, **104** may be removable from the housing **110** to separate components of the charging modules **102**, **104** (for example, charging pins, cables, circuit boards, and the like) from the housing **110**, such as for repair and/or replacement of the charging module components or other components of the charging inlet assembly **100**.
- (9) In an exemplary embodiment, the housing **110** includes an AC section **112** that receives the AC

charging module **102** and a DC section **114** that receives the DC charging module **104**. The housing **110** includes an HMI section **115** separate from the AC section **112** and the DC section **114**. In the illustrated embodiment, the HMI section **115** is located to a first side of the AC section **112** and the DC section **114**. For example, the HMI section **115** is between a left side of the housing **110** and the AC/DC sections **112**, **114**. The HMI assembly **106** is coupled to the housing **110** at the HMI section **115**. The HMI section **115** includes indicators, inputs, buttons, or other components for interface by the operator. The AC section **112** is configured for mating with an AC charging connector or an AC section of the charging connector. The DC section **114** is configured for mating with a DC charging connector or a DC section of the charging connector. The AC section **112** includes AC terminal channels **116**. The DC section **114** includes DC terminal channels **118**.

(10) The charging inlet assembly **100** includes AC terminals **120** at the AC section **112**. The AC terminals **120** are held by the housing **110**. The AC terminals **120** are received in corresponding AC terminal channels **116**. In various embodiments, the AC terminals **120** may be part of the AC charging module **102** that is coupled to the housing **110**. In the illustrated embodiment, five AC terminals **120** are provided, including a first AC charging terminal **120a**, a second AC charging terminal **120b**, a ground terminal **120c**, a proximity terminal **120d**, and a pilot terminal **120e**. Optionally, the AC terminals **120** may be different sized terminals. In an exemplary embodiment, the AC terminals **120** includes pins at mating ends of the AC terminals **120**. AC cables **122** are terminated to the AC terminals **120** and extend from the charging inlet assembly **100** to another component of the vehicle, such as the battery system of the vehicle. The AC terminals **120** are configured to be mated to the charging connector.

(11) In an exemplary embodiment, the AC section **112** of the charging inlet assembly **100** defines a low-voltage connector configured to be coupled to the low-voltage portion of the charging connectors. The low-voltage connector (for example, the AC terminals **120** and the AC cables **122**) is configured to be coupled to other components in the system, such as a battery distribution unit, to control charging of the vehicle. The low-voltage connector may transmit/receive signals relating to charging, such as status of connection, status of charge, voltage of charge, and the like. The low-voltage connector may be socket connector configured to receive the charging plug. Seals may be provided at the interface of the low-voltage connector.

(12) The charging inlet assembly **100** includes DC terminals **126** at the DC section **114**. The DC terminals **126** are held by the housing **110**. The DC terminals **126** are received in corresponding DC terminal channels **118**. In various embodiments, the DC terminals **126** may be part of the DC charging module **104** that is coupled to the housing **110**. In the illustrated embodiment, two DC terminals **126** are provided. In an exemplary embodiment, the DC terminals **126** includes pins at mating ends of the DC terminals **126**. DC cables **128** are terminated to the DC terminals **126** and extend from the charging inlet assembly **100** to another component of the vehicle, such as the battery system of the vehicle. The DC terminals **126** are configured to be mated to the charging connector.

(13) In an exemplary embodiment, the DC section **114** of the charging inlet assembly **100** defines a high-voltage connector configured to be coupled to the high-voltage portion of the charging connector. The high-voltage connector (for example, the DC terminals **126** and the DC cables **128**) is configured to be coupled to other components in the system, such as the battery and/or the battery distribution unit of the vehicle. The high-voltage connector is used for fast charging of the battery. The high-voltage connector may be socket connector configured to receive the charging plug. Seals may be provided at the interface of the high-voltage connector.

(14) The housing **110** includes a front **130** and a rear **132**. The front **130** of the housing **110** faces outward and is presented to the operator to connect the charging connector. The rear **132** faces the interior of the vehicle and is generally inaccessible without removing the housing **110** from the vehicle. In an exemplary embodiment, the housing **110** includes a panel **134** at the front **130**. In an exemplary embodiment, an AC socket **136** is formed in the panel **134** at the AC section **112** and a

DC socket **138** is formed in the panel **134** at the DC section **114**. The AC socket **136** includes a space around the AC terminals **120** that receives the charging connector. During charging, the AC charging connector is plugged into the AC socket **136** to electrically connect to the AC terminals **120**. The DC socket **138** includes a space around the DC terminals **126** that receive the charging connector. The DC charging connector is configured to be plugged into the DC socket **138**. During charging, the DC charging connector is plugged into the DC socket **138** to electrically connect to the DC terminals **116**. The panel **134** may surround the AC socket **136** and the DC socket **138** at the front **130**.

(15) In an exemplary embodiment, the housing **110** includes a rear cavity **150** at the rear **132**. In the illustrated embodiment, the rear cavity **150** is provided at the HMI section **115**. The rear cavity **150** receives the HMI assembly **106**. In an exemplary embodiment, a portion of the proximity resistor assembly **200** received in the rear cavity **150**. The rear cavity **150** is located rearward of the panel **134**. The rear cavity **150** is located within the envelope (for example, outer periphery, such as side-to-side and top-to-bottom) of the housing **110**. In the illustrated embodiment, the rear cavity **150** is open at the rear to receive the HMI assembly **106** through the opening at the rear. In an exemplary embodiment, the housing **110** includes a housing securing element (not shown) used to secure the HMI assembly **106** in the rear cavity **150**, such as a latching element, threaded fasteners, clips or other securing elements.

(16) In an exemplary embodiment, the housing **110** includes indicator openings **152** at the front **130**. The indicator openings **152** pass through the panel **134**. The HMI assembly **106** includes indicator lights at the indicator openings **152** that are visible from the front **130** of the housing **110**, such as to indicate a charging status of the charging inlet assembly **100**. In an exemplary embodiment, the HMI assembly **106** includes an actuator unlock button **154** at the front **130**. The housing **110** including an opening **156** that receives the actuator unlock button **154**. The actuator unlock button **154** is used for unlocking a latching actuator **158**, which is used for latchably securing the charging connector to the housing **110**.

(17) In an exemplary embodiment, the proximity resistor assembly **200** provides a proximity circuit with a resistor between the proximity terminal **120d** and the vehicle or charging control system, such as for operating a signaling protocol for performing the charging operation. In an exemplary embodiment, the resistor of the proximity resistor assembly **200** is included with the HMI assembly **106**. The charging inlet assembly **100** does not include a separate circuit or circuit board housing the resistor, but rather, the resistor and proximity circuit is provided as part of the HMI assembly **106**.

(18) In an exemplary embodiment, the housing **110** includes mounting tabs **140** used for mounting the housing **110** to the vehicle. The mounting tabs **140** having openings **142** that receive fasteners (not shown) to secure the charging inlet assembly **100** to the vehicle. Other types of mounting features may be used to secure the charging inlet assembly **100** to the vehicle. In various embodiments, the charging inlet assembly **100** may include a mounting flange (not shown) coupled to the front of the housing **110** for mounting the charging inlet assembly **100** to the vehicle. The housing **110** and/or the mounting flange may include a seal (not shown) to seal the charging inlet assembly **100** to the vehicle. In various embodiments, the charging inlet assembly **100** may include a terminal cover (not shown) at the front of the housing **110** to cover portions of the housing **110**, such as the DC section **114** and/or the AC section **112**. The housing **110** may include one or more rear covers at the rear of the housing **110** to close access to the rear of the housing **110**. The cover(s) may be clipped or latched onto the main part of the housing **110**, such as using clips or latches. Other types of securing features, such as fasteners may be used in alternative embodiments.

(19) FIG. 4 is a rear perspective view of a portion of the charging inlet assembly **100** showing the proximity resistor assembly **200** and the human machine interface (HMI) assembly **106** in accordance with an exemplary embodiment. FIG. 5 is a rear perspective view of a portion of the charging inlet assembly **100** showing the proximity resistor assembly **200** and a portion of the

human machine interface (HMI) assembly **106** in accordance with an exemplary embodiment. FIGS. **4** and **5** show the proximity resistor assembly **200** connected to the proximity terminal **120d**.

(20) The proximity terminal **120d** includes a main body **160** extending between a mating end **162** and a terminating end **164**. In an exemplary embodiment, a terminal seal **165** is coupled to the main body **160**. The terminal seal **165** is configured to be sealed to the housing **110** when the AC terminal **120** is received in the AC terminal channel **116**. The AC terminal **120** includes a mating pin **166** at the mating end **162**. The mating pin **166** is presented at the front **130** of the housing **110** for mating with the charging connector. The AC terminal **120** includes a crimp barrel **168** at the terminating end **164** for electrical connection (for example, crimping) with the AC cables **122**. Other types of terminating ends may be provided in alternative embodiments, such as a weld pad.

(21) The HMI assembly **106** includes an interface housing **300**, an interface controller **302**, a lens assembly **304**, and an actuator unlock assembly **306**. The interface housing **300** holds the interface controller **302**, the lens assembly **304**, and the actuator unlock assembly **306**. In an exemplary embodiment, the interface housing **300** includes a chamber **308** that receives the interface controller **302**, the lens assembly **304**, and the actuator unlock assembly **306**. The interface housing **300** is configured to be coupled to the housing **110**, such as at the rear **132** of the housing **110**. The interface housing **300** positions the interface controller **302**, the lens assembly **304**, and the actuator unlock assembly **306** relative to the housing **110** when coupled thereto.

(22) The interface housing **300** includes walls **310** defining the chamber **308**. The interface housing **300** extends between a front **312** and a rear **314**. The interface housing **300** includes a top **316** and a bottom **318**. The interface housing **300** includes sides **320**, **322** between the front **312** and the rear **314** and between the top **316** and the bottom **318**. The interface housing **300** includes receptacles **324** at the rear **314**. The receptacles **324** may receive contacts, connectors, wires or other electrical components configured to be electrically connected to the interface controller **302**. For example, the receptacles **324** may receive a portion of the proximity resistor assembly **200**.

(23) In an exemplary embodiment, the interface controller **302** includes a control circuit board **330**, one or more charging indicators **332** coupled to the control circuit board **330**, one or more board connectors **334** coupled to the control circuit board **330**, circuit components **336** coupled to the control circuit board **330**, and an actuator controller **338** coupled to the control circuit board **330**. In an exemplary embodiment, a resistor of the proximity resistor assembly **200** is coupled to the control circuit board **330**. The board connectors **334** are configured to be connected to other components, such as corresponding AC terminals or the vehicle control system to control operation of the charging inlet assembly **100** and/or functions of the HMI assembly **106**. Each connector **334** may include a connector housing and a contact held by the connector housing and mounted to the control circuit board **330**. Control signals may be received by the connectors **334** to control operation of the HMI assembly **106**, such as to control the charging indicators **332**. The connectors **334** are aligned with the receptacles **324**, such as for mating with mating connectors or wires extending to other components of the charging inlet assembly **100**. The circuit components **336** may include capacitors, resistors, processors, LED drivers, or other types of circuit components used for controlling operation of the interface controller **302**. The actuator controller **338** is aligned with the actuator unlock button **154** and configured to be operably coupled to the actuator controller **338**. For example, when the actuator controller **338** is pressed, the actuator controller **338** is operated, such as to unlock the latching actuator **158**.

(24) In an exemplary embodiment, the charging indicators **332** include LEDs **340**. The LEDs **340** are activated to provide visual indication of a charging status of the charging inlet assembly **100**. For example, one or more of the LEDs **340** may be activated to indicate if charging is occurring (ON/OFF). One or more of the LEDs **340** may be activated to indicate if charging is delayed. One or more of the LEDs **340** may be activated to indicate the charge level (for example, low charge, medium charge, full charge, and the like).

(25) The lens assembly **304** includes a lens holder **350** holding one or more lenses **352**. The lenses

352 are configured to be aligned with the LEDs **340** to receive light from the LEDs **340**. The lens assembly **304** is received in the rear cavity **150**. In various embodiments, the lens assembly **304** may be coupled to the control circuit board **330** and loaded into the rear cavity **150** with the control circuit board **330**. The lenses **352** are configured to be aligned with corresponding indicator openings **152** to direct the light from the LEDs **340** to the indicator openings **152**.

(26) During assembly, the lens assembly **304** is loaded into the rear cavity **150**, such as through the rear of the housing **110**. The lens assembly **304** may be positioned relative to the housing **110** using locating features, such as datum surfaces, guide surfaces, or other types of locating features. The lens holder **350** may be held in the rear cavity **150** by an interference fit. During assembly, the interface controller **302** is loaded into the rear cavity **150**. The interface controller **302** is positioned in the rear cavity **150** such that the LEDs **340** are aligned with the lenses **352** of the lens assembly **304**. During assembly, the interface housing **300** is coupled to the housing **110** to cover the interface controller **302**. A portion of the interface housing **300** may be loaded into the rear cavity **150**. In alternative embodiments, the lens assembly **304** and/or the interface controller **302** may be loaded into the interface housing **300** prior to coupling to the housing **110**. Once assembled, the interface housing **300** protects the interface controller **302** and the lens assembly **304** and/or retains the interface controller **302** and the lens assembly **304** in the rear cavity **150**. Once assembled, the HMI assembly **106** is fixed to the housing **110** and is secured to the vehicle with the housing **110**.

(27) In an exemplary embodiment, the proximity resistor assembly **200** includes a resistor **202** and a cable connector **204**. The proximity resistor assembly **200** may additionally include a vehicle cable connector **206**. The resistor **202** is coupled to the control circuit board **330**. For example, the resistor **202** may be coupled to a proximity circuit **208** of the control circuit board **330**. The proximity circuit **208** may be defined by one or more traces and/or circuit components of the control circuit board **330**. In an exemplary embodiment, the proximity circuit **208** is connected to a circuit ground, such as a ground plane of the control circuit board **330**, through the resistor **202**. In various embodiments, the resistor **202** may be soldered to the proximity circuit **208**.

(28) The cable connector **204** includes a connector **210** and a cable **212** extending between the connector **210** and the proximity terminal **120d**. The connector **210** includes a connector housing **214** and a contact **216** held by the connector housing **214**. The contact **216** is terminated to an end of the cable **212**. For example, the contact **216** may be soldered to the end of the cable **212** or crimped to the end of the cable **212**. The other end of the cable **212** may be soldered or crimped to the terminating end of the proximity terminal **120d**. The connector **210** is configured to be connected with the board connector **334**. For example, the connector **210** may be a plug connector and the board connector **334** may be a receptacle connector configured to receive the connector **210**, or vice versa. The connector **210** is configured to be plugged into the receptacle **324** to mate with the board connector **334**. In an exemplary embodiment, the connector **210** includes a perimeter seal **218** around the outer perimeter of the connector housing **214**. The perimeter seal **218** is configured to be sealed to the interface housing **300** when the connector **210** is plugged into the receptacle **324**. When mated, the contact **216** is electrically connected to a contact of the board connector **334** to electrically connect the cable connector **204** to the control circuit board **330**, such as to the proximity circuit **208**. The cable **212** transmits proximity signals from the proximity terminal **120d** to the proximity circuit **208** through the connectors **210**, **334**.

(29) In an exemplary embodiment, the proximity resistor assembly **200** includes a cable seal **213** coupled to the cable **212**. The cable seal **213** is configured to be sealed against the cable **212**. The cable seal **213** may be provided proximate to the end of the cable **212** that is terminated to the proximity terminal **120d**. In an exemplary embodiment, the cable seal **213** is received in the DC terminal channel **118** to seal against the housing **110**.

(30) The vehicle cable connector **206** includes a connector **220** and a cable **222** extending between the connector **220** and a vehicle electronics module **230**. The vehicle electronics module **230** may be used to control a charging operation of the charging inlet assembly **100**. The vehicle electronics

module **230** may be part of the battery distribution unit. The connector **220** includes a connector housing **224** and a contact **226** held by the connector housing **224**. The contact **226** is terminated to an end of the cable **222**. For example, the contact **226** may be soldered to the end of the cable **222** or crimped to the end of the cable **222**. The other end of the cable **222** may be soldered to the vehicle electronics module **230** or connected to a connector coupled to the vehicle electronics module **230**. Proximity signals are transmitted from the proximity terminal **120d** to the vehicle electronics module **230** through the cable connector **204**, through the proximity circuit **208** on the control circuit board **330** of the HMI assembly **106**, and through the vehicle cable connector **206**. As such, the proximity signal passes through the resistor **202** prior to transmission through the vehicle cable connector **206** to the vehicle electronics module **230**. The connector **220** is configured to be connected with the corresponding board connector **334**. For example, the connector **220** may be plugged into the receptacle **324** to mate with the board connector **334**. When mated, the contact **226** is electrically connected to a contact of the board connector **334** to electrically connect the cable connector **204** to the control circuit board **330**, such as to the proximity circuit **208**. The cable **222** transmits proximity signals from the proximity circuit **208** to the vehicle electronics module **230**.

(31) The resistor **202** and the proximity circuit **208** are used to satisfy signaling requirements for the charging inlet assembly **100**. For example, in an exemplary embodiment, the resistor **202** is a 2.7 kOhm resistor, which is required for operation to satisfy standards for charging system for electric vehicles, such as the SAE J1772 North American Standard. The resistor **202** and proximity circuit **208** are provided on the control circuit board **330** of the HMI assembly **106**. Providing the resistor **202** and proximity circuit **208** on the control circuit board **330** of the HMI assembly **106** eliminates the need for an additional printed circuit board or other circuitry at the AC section **112**, which may reduce the overall size of the charging inlet assembly **100** compared to charging inlet assemblies having separate control circuit boards and housing spaces for such control circuit boards. The cable connector **204** provides a reliable and convenient connection between the proximity terminal **120d** and the HMI assembly **106**. Assembly can be made quickly and easily.

(32) It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from its scope.

Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments, and are by no means limiting and are merely exemplary embodiments. Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

Claims

1. A charging inlet assembly comprising: a housing extending between a front and a rear, the housing having a DC section including DC terminal channels, the housing having an AC section including AC terminal channels, the DC section configured for mating with a DC charging

connector at the front, the AC section configured for mating with an AC charging connector at the front; DC terminals received in the corresponding DC terminal channels for mating with the DC charging connector, the DC terminals including a proximity terminal; AC terminals received in the corresponding AC terminal channels for mating with the AC charging connector; and a human machine interface (HMI) assembly coupled to the rear of the housing, the HMI assembly including a control circuit board, the HMI assembly including a charging indicator coupled to the control circuit board, the charging indicator being visible from the front of the housing to indicate a charging status of the charging inlet assembly; and a proximity resistor assembly coupled to the housing, the proximity resistor assembly including a resistor coupled to the control circuit board, the proximity resistor assembly including a cable connector connected between the control circuit board and the proximity terminal.

2. The charging inlet assembly of claim 1, wherein the cable connector includes a cable and a contact at a first end of the cable, the contact being electrically connected to the control circuit board, a second end of the cable being terminated to the proximity terminal.
3. The charging inlet assembly of claim 1, wherein the cable connector transmits proximity signals between the proximity terminal and the control circuit board.
4. The charging inlet assembly of claim 3, wherein the proximity signals are used to control operation of the charging indicator.
5. The charging inlet assembly of claim 1, wherein the resistor is connected to a proximity circuit of the control circuit board, the proximity resistor assembly further comprising a vehicle cable connector connected between the proximity circuit of the control circuit board and a vehicle electronics module used to control a charging operation of the charging inlet assembly.
6. The charging inlet assembly of claim 1, wherein the housing includes an HMI section discrete from the DC section and the AC section, the HMI assembly coupled to the housing at the HMI section.
7. The charging inlet assembly of claim 6, wherein the housing includes a rear cavity at the HMI section, the HMI assembly being received in the rear cavity, the resistor being located in the rear cavity.
8. The charging inlet assembly of claim 6, wherein the HMI section is offset toward a first side of the housing from the DC section and the AC section.
9. The charging inlet assembly of claim 1, wherein the control circuit board includes a front edge and a rear edge, the control circuit board having a mounting surface between the front edge and the rear edge, the resistor being coupled to the mounting surface, the control circuit board being coupled to the housing with the front edge being forward facing and the rear edge being rearward facing.
10. The charging inlet assembly of claim 1, wherein the HMI assembly includes an interface housing having a chamber, the chamber receives the control circuit board, the cable connector being connected to the interface housing.
11. The charging inlet assembly of claim 10, wherein the interface housing includes a receptacle, the HMI assembly including a board connector mounted to the control circuit board, the board connector being located in the receptacle, the cable connector being plugged into the receptacle to mate with the board connector.
12. The charging inlet assembly of claim 1, wherein the cable connector includes a cable, the proximity resistor assembly including a cable seal coupled to the cable, the cable seal being received in the DC terminal channel to seal against the housing.
13. A charging inlet assembly comprising: a housing extending between a front and a rear, the housing having a DC section including DC terminal channels, the housing having an AC section including AC terminal channels, the housing having a human machine interface (HMI) section discrete from the DC section and the AC section, the DC section configured for mating with a DC charging connector at the front, the AC section configured for mating with an AC charging

connector at the front, the housing including a rear cavity at the HMI section, the rear cavity being separate from the DC terminal channels and from the AC terminal channels; DC terminals received in the corresponding DC terminal channels for mating with the DC charging connector, the DC terminals including a proximity terminal; AC terminals received in the corresponding AC terminal channels for mating with the AC charging connector; and an HMI assembly received in the rear cavity, the HMI assembly including a control circuit board, the HMI assembly including a charging indicator coupled to the control circuit board, the charging indicator located within the HMI section and being visible from the front of the housing to indicate a charging status of the charging inlet assembly; and a proximity resistor assembly coupled to the housing, the proximity resistor assembly including a resistor coupled to the control circuit board, the proximity resistor assembly including a cable connector connected between the control circuit board and the proximity terminal.

14. The charging inlet assembly of claim 13, wherein the HMI section is offset toward a first side of the housing from the DC section and the AC section.

15. The charging inlet assembly of claim 13, wherein the cable connector includes a cable and a contact at a first end of the cable, the contact being electrically connected to the control circuit board, a second end of the cable being terminated to the proximity terminal.

16. The charging inlet assembly of claim 13, wherein the cable connector transmits proximity signals between the proximity terminal and the control circuit board, the proximity signals being used to control operation of the charging indicator.

17. The charging inlet assembly of claim 13, wherein the HMI assembly includes an interface housing having a chamber, the chamber receives the control circuit board, the interface housing includes a receptacle, the HMI assembly including a board connector mounted to the control circuit board, the board connector being located in the receptacle, the cable connector being plugged into the receptacle to mate with the board connector.

18. The charging inlet assembly of claim 13, wherein the cable connector includes a cable, the proximity resistor assembly including a cable seal coupled to the cable, the cable seal being received in the DC terminal channel to seal against the housing.

19. A charging inlet assembly comprising: a housing extending between a front and a rear, the housing having a DC section including DC terminal channels, the housing having an AC section including AC terminal channels, the DC section configured for mating with a DC charging connector at the front, the AC section configured for mating with an AC charging connector at the front; DC terminals received in the corresponding DC terminal channels for mating with the DC charging connector, the DC terminals including a proximity terminal; AC terminals received in the corresponding AC terminal channels for mating with the AC charging connector; and a human machine interface (HMI) assembly coupled to the rear of the housing, the HMI assembly including a control circuit board, the HMI assembly including a charging indicator coupled to the control circuit board, the charging indicator being visible from the front of the housing to indicate a charging status of the charging inlet assembly; and a proximity resistor assembly coupled to the housing, the proximity resistor assembly including a resistor and a cable connector, the resistor coupled to a proximity circuit of the control circuit board, the resistor coupled to a ground of the control circuit board, the cable connector including a contact connected to the proximity circuit of the control circuit board, the cable connector including a cable connected between the contact and the proximity terminal to receive a proximity signal from the proximity terminal.

20. The charging inlet assembly of claim 19, wherein the housing includes an HMI section discrete from the DC section and the AC section, the HMI assembly coupled to the housing at the HMI section, the HMI section is offset toward a first side of the housing from the DC section and the AC section.
